NORMAL ORDERS OF THE ERDŐS CHAIN FUNCTIONS $h(n)$ AND $H(n)$

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Abstract. For a positive integer n let $h(n)$ be the largest ℓ such that there exist primes $p_1 < \cdots < p_\ell$ all dividing n with $p_{i+1} \equiv 1 \pmod{p_i}$, and let $H(n)$ be the analogous quantity in which the p_i are replaced by arbitrary divisors $d_i \mid n$. Erdős asked [Erd] for estimates of $h(n)$ and $H(n)$, and whether $H(n)/h(n) \to \infty$ for almost all n; he suggested, tentatively, that the normal order of $h(n)$ should be about $\log_* n$.

We prove that for all but $o(x)$ integers $n \le x$,

$$h(n) = \Big(\tfrac{1}{2} + o(1)\Big) \log_* n, \qquad H(n) = \big(1 + o(1)\big) \log_* n, \qquad \frac{H(n)}{h(n)} = 2 + o(1).$$

In particular $H(n)/h(n) \to \infty$ is *false* on a density-one set, while Erdős's prediction for the scale of $h(n)$ is confirmed: $h(n) = \Theta(\log_* n)$ almost always, with leading constant $1/2$.

The upper bounds use a bad-pair density argument combined with an iterated-tower counting lemma. The lower bound for h is a Siegel–Walfisz prime-successor construction transferred to integers via the Chinese remainder theorem. The lower bound for H uses a new *subset-product successor lemma*: if $g_1, \ldots, g_K$ are independent uniformly distributed elements of a finite abelian group G of order N, then with probability at least $1 - N/(2^K - 1)$ some nonempty subproduct of the g_i is equal to the identity. This is applied, after a near-uniform residue reduction using Siegel–Walfisz, inside Mertens-sized prime windows.

1. Introduction

Let $n \ge 2$ be an integer. Erdős introduced two natural "chain" functions attached to the multiplicative structure of n:

(i) $h(n)$ is the largest integer ℓ such that there exist primes $p_1 < p_2 < \cdots < p_\ell$ all dividing n, with $p_{i+1} \equiv 1 \pmod{p_i}$ for $1 \le i < \ell$;

(ii) $H(n)$ is the largest integer u such that there exist integers $d_1 < d_2 < \cdots < d_u$ all dividing n, with $d_{i+1} \equiv 1 \pmod{d_i}$ for $1 \le i < u$.

We set $h(1) = H(1) = 0$. For $n \ge 2$, $h(n) \ge 1$ because n has a prime divisor, and $H(n) \ge 1$ because the one-term divisor chain $1 \mid n$ is allowed.

Problem 696 of the Erdős Problems database [Erd] asks: estimate $h(n)$ and $H(n)$, and decide whether

$$\frac{H(n)}{h(n)} \longrightarrow \infty \qquad \text{for almost all } n. \tag{1}$$

According to the Erdős Problems Database [Erd], Erdős further suggested that the *normal order* of $h(n)$ should be the iterated logarithm $\log_* n$. (We were unable to locate this conjecture in Erdős's published papers; the reader should treat the attribution as based on the database commentary.)

The first nontrivial result on this problem (folklore; see van Doorn's comments in [Erd]) is that $h(n) \to \infty$ for almost all n. The present paper resolves the full problem by determining the normal orders of both h and H *with explicit constants*.

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1.1. **Statement of the result.** Define $T_0 := e$ and $T_{m+1} := \exp(T_m)$ (these are formally introduced in Section 2). We set

$$\log_* x \;:=\; \min\{m \geq 0 : x \leq T_m\}. \tag{2}$$

Equivalently, for $x \geq e$, this is

$$\log_* x \;=\; \min\{m \geq 0 : \log^{(m)} x \leq e\}.$$

Thus $\log_* e = 0$, $\log_*(e^e) = 1$, etc. This convention differs from some other conventions by at most 1, which is immaterial for the normal-order statements below.

Theorem 1.1 (Main theorem)**.** *For all but $o(x)$ integers $n \leq x$,*

$$h(n) = \Big(\tfrac{1}{2} + o(1)\Big) \log_* n, \tag{3}$$

$$H(n) = \big(1 + o(1)\big) \log_* n, \tag{4}$$

$$\frac{H(n)}{h(n)} = 2 + o(1). \tag{5}$$

In particular, the divergence (1) is false on a density-one set: $H(n)/h(n)$ is bounded above by $2 + o(1)$ for almost all n.

It is striking that the constant $1/2$ for h and the constant 1 for H both appear, and that the ratio is exactly 2. The factor of 2 has a transparent combinatorial explanation: a divisor-chain step $d_i \to d_{i+1}$ with $d_{i+1} \equiv 1 \pmod{d_i}$ allows d_{i+1} to be any product of selected primes hitting the trivial residue class mod d_i, whereas a prime-chain step $p_i \to p_{i+1}$ requires p_{i+1} itself to be a prime in the residue class $1 \pmod{p_i}$. For divisor chains the cost of one tower step is essentially $\log_2 d_i$ random primes; for prime chains the cost is essentially d_i itself, and the bad-pair argument shows that two prime steps lower the iterated-tower rank by exactly two when one would have been lowered by one for a single divisor step.

1.2. **Comparison with what was previously known.** The qualitative claim $h(n) \to \infty$ a.a. was known. The quantitative statement $h(n) = \Theta(\log_* n)$ a.a. does not appear to have been published formally, although the proof we give in Section 4 for the upper bound is essentially classical, drawing on Brun–Titchmarsh and Mertens, and the Θ-form lower bound follows the strategy implicit in Erdős–Granville–Pomerance–Spiro [EGPS90] on iterates of the totient function.

The closest published work in print is Ford, Konyagin and Luca [FKL10], who study the Pratt-tree depth

$$H_{\mathrm{Pr}}(p) \;:=\; \max\big\{k : \exists\, p_1 \prec p_2 \prec \cdots \prec p_k = p\big\},$$

where each p_i is prime and $a \prec b$ means $b \equiv 1 \pmod{a}$. They prove $H_{\mathrm{Pr}}(p) \leq (\log p)^{1-c'}$ for almost all primes $p \leq x$ (some $c' > 0$), and conjecture that the normal order is $e \log\log x$ via a branching random walk model. *This is a different problem.* In our setup, the chain primes/divisors must all divide a specified integer n, which is a much more restrictive constraint than "terminating at a fixed prime p". The Pratt-tree depth grows like $\log\log p$; our $h(n)$ grows like $\frac{1}{2}\log_* n$, much slower, because the supply of chain primes is restricted to the $\sim \log\log n$ prime divisors of n.

The sharp normal orders (3)–(4) (with the matching constants $1/2$ and 1) and the resulting refutation (5) of (1), are, to the best of the authors' knowledge, new. The upper bound for h in (3) is a Brun–Titchmarsh refinement of the bad-pair argument; the matching lower bound is built greedily on the Siegel–Walfisz prime-counting function in arithmetic progressions. The lower bound (4) for H rests on a probabilistic *subset-product successor lemma*, stated and proved in Section 6; this is the new technical input of the paper.

1.3. **Acknowledgement of AI assistance.** The arguments in this paper were assembled in collaboration with the Anthropic language model Claude Opus, working from a four-turn proof sketch produced by GPT-5.5-Pro (extended) on April 26, 2026. The authors verified all displayed inequalities and, in several places, used GPT-5.5-Pro (extended thinking) to fill in arguments that the original sketch had recorded only schematically. The result statement and overall proof structure are due to GPT-5.5-Pro; the iterative patches refining the writeup are also due to GPT-5.5-Pro (extended), assembled together with Claude Code.

1.4. **Organization.** Section 2 fixes notation and recalls the analytic inputs. Section 3 proves the upper bound $H(n) \le (1+o(1)) \log_* n$ a.a. Section 4 proves the upper bound $h(n) \le (\frac{1}{2}+o(1)) \log_* n$ a.a. Section 5 proves the matching lower bound $h(n) \ge (\frac{1}{2} - o(1)) \log_* n$ a.a. via Siegel–Walfisz prime mass. Section 6 establishes the subset-product successor lemma. Section 7 proves $H(n) \ge (1 - o(1)) \log_* n$ a.a. using that lemma. Section 8 assembles the main theorem. Section 9 contains concluding remarks.

2. Notation, conventions, and analytic inputs

2.1. **Asymptotic conventions.** Throughout the paper x is a real parameter tending to ∞. We write $f \ll g$ or $f = O(g)$ to mean that $|f| \le C|g|$ for some absolute constant $C > 0$ and all sufficiently large x, and $f = o(g)$ if $f/g \to 0$ as $x \to \infty$. The implicit constants in any O or $\ll$ may depend only on absolute parameters introduced *before* the relevant inequality is asserted, and are explicitly labeled where needed.

We say a property $\mathcal{P} = \mathcal{P}(n)$ holds *for almost all* n (equivalently, "a.a."), or that $\mathcal{P}$ holds with *density one*, if

$$\#\big\{n \le x : \neg\mathcal{P}(n)\big\} \;=\; o(x) \qquad (x \to \infty). \tag{6}$$

2.2. **Iterated logarithms and towers.** All logarithms in the paper are natural logarithms. We use $\log^{(m)} x$ for the m-fold iterated logarithm, with $\log^{(0)} x = x$.

We use the iterated logarithm $\log_* x$ as in (2). Equivalently, the tower of base-e exponentials is defined by

$$T_0 \;=\; e, \qquad T_{m+1} \;=\; \exp(T_m) \qquad (m \ge 0), \tag{7}$$

and $\log_* x \le m$ if and only if $x \le T_m$. Note the basic identity

$$\log T_m \;=\; T_{m-1} \qquad (m \ge 1). \tag{8}$$

We will use the auxiliary scale

$$U_m \;:=\; T_m^3 \qquad (m \ge 0), \tag{9}$$

which behaves comparably to the tower T_m but provides slack for constant-factor losses.

We record several elementary computations that are used repeatedly:

$$\log U_m = 3T_{m-1}, \tag{10}$$

$$\log\log U_m = \log\big(3T_{m-1}\big) = T_{m-2} + \log 3, \tag{11}$$

$$\log\log\log U_m = \log\big(T_{m-2} + \log 3\big) = T_{m-3} + O\big(T_{m-3}^{-1}\big). \tag{12}$$

Identities (10)–(12) are valid for $m \ge 4$, by repeated use of (8).

2.3. **Standard analytic inputs.** We will appeal to four classical theorems of analytic number theory; we quote them in the precise form used.

Lemma 2.1 (Siegel–Walfisz, [Dav, Ch. 22], [SW])**.** *For every $A_0 > 0$ there exists $c_0 = c_0(A_0) > 0$ such that, uniformly for $t \geq 3$, $q \leq (\log t)^{A_0}$, and $(a, q) = 1$,*

$$\pi(t; q, a) \;=\; \frac{\mathrm{Li}(t)}{\varphi(q)} + O_{A_0}\Big(t \exp\big(-c_0\sqrt{\log t}\big)\Big). \tag{13}$$

Consequently, if $3 \leq X \leq Y$, $q \leq (\log X)^{A_0}$, and $(a, q) = 1$, then

$$\sum_{\substack{X<p\leq Y\\ p\equiv a \pmod q}} \frac{1}{p} \;=\; \frac{1}{\varphi(q)} \int_X^Y \frac{dt}{t \log t} + O_{A_0}\Big(\exp\big(-c_0\sqrt{\log X}\big)\Big), \tag{14}$$

after possibly decreasing c_0. The implied constants depend only on A_0.

Proof of (14) *given* (13). Since $q \leq (\log X)^{A_0}$ and $t \geq X$, (13) is valid uniformly throughout $[X, Y]$. Partial summation gives

$$\sum_{\substack{X<p\leq Y\\ p\equiv a \pmod q}} \frac{1}{p} \;=\; \frac{\pi(Y; q, a)}{Y} - \frac{\pi(X; q, a)}{X} + \int_X^Y \frac{\pi(t; q, a)}{t^2}\, dt.$$

Substituting (13), the main term is

$$\frac{1}{\varphi(q)}\left(\frac{\mathrm{Li}(Y)}{Y} - \frac{\mathrm{Li}(X)}{X} + \int_X^Y \frac{\mathrm{Li}(t)}{t^2}\, dt\right) \;=\; \frac{1}{\varphi(q)} \int_X^Y \frac{dt}{t \log t}$$

(after partial integration of the right-hand side, using $\mathrm{Li}'(t) = 1/\log t$). The error term is

$$O_{A_0}\left(\exp(-c_0\sqrt{\log X}) + \int_X^Y \frac{\exp(-c_0\sqrt{\log t})}{t}\, dt\right).$$

With $u = \sqrt{\log t}$, the integral is

$$\ll \int_{\sqrt{\log X}}^{\infty} u\, e^{-c_0 u}\, du \;\ll\; \exp\Big(-\tfrac{c_0}{2}\sqrt{\log X}\Big),$$

so, after decreasing c_0, the total error is $O_{A_0}(\exp(-c_0\sqrt{\log X}))$. □ □

Remark 2.2 (Uniformity in the Siegel–Walfisz input)**.** The constant c_0 and all implied constants in Lemma 2.1 depend only on the fixed exponent A_0. In the sequel we use this lemma only with $A_0 = 1$; after this choice, the constants are absolute. In particular, when (14) is applied on an interval $(X, Y]$, the error term is uniform in the upper endpoint Y, because the proof bounded the error integral by the tail integral over $[X, \infty)$. The usual ineffectivity of the Siegel–Walfisz constant is irrelevant here, since all constants are fixed before $x \to \infty$ and the arguments are qualitative density-one estimates.

Remark 2.3 (Amount of Siegel–Walfisz actually needed)**.** The exponential error term in Lemma 2.1 is convenient, but the arguments below do not rely on optimizing this error. It would be enough, in all later applications, to have the following weaker consequence for one fixed large exponent B_{SW}:

$$\sum_{\substack{X<p\leq Y\\ p\equiv a \pmod q}} \frac{1}{p} = \frac{1}{\varphi(q)} \int_X^Y \frac{dt}{t \log t} + O_{A_0, B_{\mathrm{SW}}}\big((\log X)^{-B_{\mathrm{SW}}}\big), \tag{15}$$

uniformly for $3 \le X \le Y$, $q \le (\log X)^{A_0}$, and $(a,q) = 1$. Indeed, in Lemma 5.2 one has $X = \exp(\exp y)$, so the error in (15) would be $O(\exp(-B_{\mathrm{SW}} y))$, which is negligible compared with the main term there. In Lemma 6.3 one has $X = \alpha \ge \exp y$ and $d \le y$; after summing over at most $\varphi(d) \le y$ reduced residue classes, the contribution of the error term would be $O(y^{1-B_{\mathrm{SW}}})$, which is $O(y^{-1/2})$ as soon as $B_{\mathrm{SW}} \ge 3$. Thus the later proof uses only standard Siegel–Walfisz uniformity for fixed powers of $\log X$; the stronger exponential form is quoted to keep the notation clean.

Lemma 2.4 (Brun–Titchmarsh, [IK, Thm. 6.6], [MV])**.** *There is an absolute constant $C_{BT} > 0$ such that for all $q \ge 1$, all a coprime to q, and all $t \ge 2q$,*

$$\pi(t; q, a) \;\le\; \frac{C_{BT}\, t}{\varphi(q)\, \log(t/q)}. \tag{16}$$

Lemma 2.5 (Mertens, [HW, Thm. 427], [Mer])**.** *There is an absolute constant $\mathcal{M} \in \mathbb{R}$ such that*

$$\sum_{p \le t} \frac{1}{p} \;=\; \log\log t \;+\; \mathcal{M} \;+\; O\!\left(\frac{1}{\log t}\right) \qquad (t \ge 2). \tag{17}$$

Lemma 2.6 (Chebyshev, [HW, Thm. 414])**.** *There is an absolute constant $C_\theta > 0$ such that*

$$\sum_{p \le t} \log p \;\le\; C_\theta\, t \qquad (t \ge 2). \tag{18}$$

2.4. **Probabilistic primes.** Throughout the lower-bound arguments we work in the *independent prime-divisibility model*: we declare each prime p to be "selected" independently with probability $1/p$. We then transfer back to genuine integers $n \le x$ via the Chinese remainder theorem.

The transfer principle is the following standard observation, recorded once for later reuse:

Lemma 2.7 (CRT transfer)**.** *Let $P \ge 2$, $M = \prod_{p \le P} p$, and let $\mathcal{E}$ be an event depending only on the indicator vector $\big(\mathbf{1}_{p|n}\big)_{p \le P}$. Let $\mathbb{P}_{\mathrm{prod}}(\mathcal{E})$ denote the probability of $\mathcal{E}$ in the product model where the $\mathbf{1}_{p|n}$ are independent Bernoulli$(1/p)$ random variables. Then*

$$\frac{1}{x}\#\{n \le x : \mathcal{E}(n) \text{ holds}\} \;=\; \mathbb{P}_{\mathrm{prod}}(\mathcal{E}) \;+\; O\big(M/x\big). \tag{19}$$

In particular, if $M = o(x)$, the density of integers $n \le x$ for which $\mathcal{E}$ holds equals $\mathbb{P}_{\mathrm{prod}}(\mathcal{E}) + o(1)$.

Sketch. By the Chinese remainder theorem the map $n \bmod M \mapsto (n \bmod p)_{p \le P}$ is a bijection. Under the uniform distribution on $\mathbb{Z}/M\mathbb{Z}$, the events $\{p \mid n\}$ for $p \le P$ are therefore independent and each has probability $1/p$, exactly matching the product model. Hence the event $\mathcal{E}$ is a union of residue classes mod M, with measure $\mathbb{P}_{\mathrm{prod}}(\mathcal{E})$ under the uniform distribution on $\mathbb{Z}/M\mathbb{Z}$. Counting integers in $[1, x]$ in each residue class incurs an error of ± 1 per class; since there are at most M classes the total error is $O(M)$, and dividing by x gives (19). □

When the event $\mathcal{E}$ depends not only on prime divisibilities but also on whether n is divisible by some *higher* prime power p^2, the same argument applies with M replaced by $\prod_{p \le P} p^{k_p}$. We will not need this generalization.

Remark 2.8 (Squarefree products under the CRT transfer)**.** In the lower-bound applications of Lemma 2.7, the constructed chain terms are squarefree products of primes whose selected status is among the coordinates $\big(\mathbf{1}_{p|n}\big)_{p \le P}$. After transfer to an actual integer n, "selected" means $p \mid n$. Hence any squarefree product e of selected primes with all prime factors at most P satisfies $e \mid n$. In particular, whenever such an e occurs for an integer $n \le x$, one automatically has $e \le n \le x$. Moreover $e \le M = \prod_{p \le P} p$, and in both lower-bound applications below we verify $M = o(x)$;

thus all squarefree products considered in the CRT transfer are $< x$ for all sufficiently large x. No additional size condition is hidden in the passage from the product model to integers.

Lemma 2.9 (Chernoff bound for independent Bernoulli variables)**.** *Let $I_1, \dots, I_M$ be independent $\{0,1\}$-valued random variables, not necessarily identically distributed, and put*

$$W := \sum_{k=1}^{M} I_k, \qquad \mu := \mathbb{E}W.$$

Then, for every $0 < \delta < 1$,

$$\mathbb{P}\big(W \le (1-\delta)\mu\big) \le \exp\left(-\frac{\delta^2 \mu}{2}\right). \tag{20}$$

In particular,

$$\mathbb{P}\big(W \le (2/3)\mu\big) \le \exp(-\mu/18). \tag{21}$$

Proof. Write $p_k := \mathbb{E}I_k$. For $\lambda > 0$, Markov's inequality and independence give

$$\mathbb{P}(W \le (1-\delta)\mu) = \mathbb{P}\big(e^{-\lambda W} \ge e^{-\lambda(1-\delta)\mu}\big) \le e^{\lambda(1-\delta)\mu} \prod_{k=1}^{M} \big(1 - p_k + p_k e^{-\lambda}\big).$$

Using $1 + u \le e^u$,

$$\prod_{k=1}^{M} \big(1 - p_k + p_k e^{-\lambda}\big) \le \exp\left(-(1-e^{-\lambda}) \sum_{k=1}^{M} p_k\right) = \exp\Big(-(1-e^{-\lambda})\mu\Big).$$

Choose $\lambda = \log(1/(1-\delta))$, so that $1 - e^{-\lambda} = \delta$. Then

$$\mathbb{P}(W \le (1-\delta)\mu) \le \exp\left(-\mu \left[\delta - (1-\delta)\log\frac{1}{1-\delta}\right]\right).$$

The elementary inequality

$$\delta - (1-\delta)\log\frac{1}{1-\delta} \ge \frac{\delta^2}{2} \qquad (0 < \delta < 1)$$

gives (20). Taking $\delta = 1/3$ gives (21). □

Remark 2.10 (CRT transfer for x-dependent events)**.** Lemma 2.7 is applied below to events $\mathcal{E} = \mathcal{E}_x$ and cutoffs $P = P(x)$ that vary with x. This causes no additional difficulty: for each fixed x, the lemma is an exact finite CRT statement with error $O(M/x)$, where $M = \prod_{p \le P} p$. Thus it is enough, in each lower-bound application, to verify $M = o(x)$ for the particular cutoff used there.

In practice we verify the stronger condition $P = o(\log x)$. By Lemma 2.6,

$$\log M = \sum_{p \le P} \log p \le C_\theta P = o(\log x),$$

and therefore $M = \exp(o(\log x)) = x^{o(1)} = o(x)$. The deterministic rules used later, such as choosing the least selected prime or the least admissible squarefree product, ensure that the relevant events are genuine functions of the finite indicator vector

$$\big(\mathbf{1}_{p|n}\big)_{p \le P}.$$

No information about primes above P or about higher prime powers is used in those CRT transfers.

3. Upper bound for $H(n)$

In this section we prove:

Theorem 3.1. *For all but $o(x)$ integers $n \le x$,*

$$H(n) \;\le\; \big(1+o(1)\big)\log_* x. \tag{22}$$

The strategy: any divisor chain step $d_i < d_{i+1}$ with $d_{i+1} \equiv 1 \pmod{d_i}$ that does not satisfy $d_{i+1} \ge \exp(\sqrt{d_i})$ creates a "bad pair" (d_i, d_{i+1}). We show that bad pairs of moderate size are rare on average, so for almost all n no large step is bad, forcing each chain step to climb the iterated tower by roughly one full level.

3.1. The bad-pair lemma.

Definition 3.2 (Bad pair, divisor version)**.** An ordered pair of positive integers (d, e) is called *H-bad* if

$$d < e, \qquad e \equiv 1 \pmod{d}, \qquad e < \exp\big(\sqrt{d}\big). \tag{23}$$

Lemma 3.3. *Let $Y = Y(x) \to \infty$. The number of integers $n \le x$ such that some H-bad pair (d, e) with $d \ge Y$ has both $d \mid n$ and $e \mid n$ is at most*

$$O\Big(x\,Y^{-1/2}\Big) \;=\; o(x). \tag{24}$$

Proof. The congruence $e \equiv 1 \pmod{d}$ forces $\gcd(d, e) = 1$, hence $\operatorname{lcm}(d, e) = de$ and the number of $n \le x$ divisible by both d and e is $\lfloor x/(de) \rfloor \le x/(de)$. The total count is therefore bounded by

$$\sum_{d \ge Y} \frac{x}{d} \sum_{\substack{d<e<\exp(\sqrt{d}) \\ e\equiv 1 \pmod{d}}} \frac{1}{e}. \tag{25}$$

For fixed $d \ge 4$, write $e = ad + 1$ with $1 \le a \le \exp(\sqrt{d})/d$:

$$\sum_{\substack{d<e<\exp(\sqrt{d}) \\ e\equiv 1 \pmod{d}}} \frac{1}{e} \;\le\; \sum_{1\le a\le \exp(\sqrt{d})/d} \frac{1}{ad+1} \;\le\; \frac{1}{d} \sum_{1\le a\le \exp(\sqrt{d})/d} \frac{1}{a}. \tag{26}$$

The harmonic sum is $\le 1 + \log\big(\exp(\sqrt{d})/d\big) \le \sqrt{d}$, valid for $d \ge 4$. Substituting,

$$\sum_{\substack{d<e<\exp(\sqrt{d}) \\ e\equiv 1 \pmod{d}}} \frac{1}{e} \;\le\; \frac{\sqrt{d}}{d} \;=\; d^{-1/2}. \tag{27}$$

Inserting (27) into (25),

$$\sum_{d \ge Y} \frac{x}{d} \cdot d^{-1/2} \;=\; x \sum_{d \ge Y} d^{-3/2} \;\ll\; x\,Y^{-1/2}. \tag{28}$$

Choosing $Y \to \infty$ gives (24). □

3.2. The U-tower lemma for H.

Set

$$G(t) \;:=\; (\log t)^2. \tag{29}$$

Lemma 3.4. *For every $m \ge 5$,*

$$G(U_m) \;\le\; U_{m-1}. \tag{30}$$

Proof. By (10), $\log U_m = 3T_{m-1}$, hence

$$G(U_m) \;=\; (\log U_m)^2 \;=\; 9T_{m-1}^2.$$

We must show $9T_{m-1}^2 \le T_{m-1}^3 = U_{m-1}$, equivalently $T_{m-1} \ge 9$. Since $T_4 = \exp\bigl(\exp\bigl(\exp(e)\bigr)\bigr)$ is enormous, this holds whenever $m-1 \ge 1$, i.e. $m \ge 2$; we asked $m \ge 5$ to absorb the $\log 3$ in (11) for any later use, but it is not needed for (30). □

3.3. **Proof of Theorem 3.1.** Let x be large, $L := \log_* x$, and put $Y := \lfloor L^{1/2} \rfloor$. By Lemma 3.3, all but $o(x)$ integers $n \le x$ have the property

$$\text{no } H\text{-bad pair } (d,e) \text{ with } d \ge Y \text{ has both } d \mid n \text{ and } e \mid n. \tag{$\star_H$}$$

Fix such an n and let $d_1 < d_2 < \cdots < d_u$ be a divisor chain witnessing $H(n) = u$. Let $u_{\text{small}} := \#\{i : d_i < Y\}$ and $u_{\text{large}} := u - u_{\text{small}}$.

Trivially $u_{\text{small}} \le Y$, since the small terms are distinct integers in $[1, Y)$.

For each large index i with $d_i \ge Y$ and $i < u$, the pair (d_i, d_{i+1}) satisfies $d_i < d_{i+1}$, $d_{i+1} \equiv 1 \pmod{d_i}$, both divide n, and $d_i \ge Y$. Property $(\star_H)$ rules out $d_{i+1} < \exp(\sqrt{d_i})$, so

$$d_{i+1} \;\ge\; \exp\bigl(\sqrt{d_i}\bigr), \qquad \text{equivalently,} \qquad d_i \;\le\; (\log d_{i+1})^2 \;=\; G(d_{i+1}). \tag{31}$$

Define the *U-rank* of d to be $\rho(d) := \min\{m \ge 0 : d \le U_m\}$. Since $d_u \le n \le x \le T_L \le U_L$, we have $\rho(d_u) \le L$.

Lemma 3.5 (Tower descent for H)**.** *If $d_i \ge Y$ and $i < u$ then $\rho(d_i) \le \rho(d_{i+1}) - 1$, provided $\rho(d_{i+1}) \ge 6$.*

Proof. Set $m := \rho(d_{i+1})$. Then $d_{i+1} \le U_m$, so by (31) and Lemma 3.4,

$$d_i \;\le\; G(d_{i+1}) \;\le\; G(U_m) \;\le\; U_{m-1}, \tag{32}$$

provided $m \ge 5$ so that $U_m \ge T_5 \ge 9$. Hence $\rho(d_i) \le m - 1 = \rho(d_{i+1}) - 1$. □

For completeness, we spell out the rank count used in the next paragraph. The terms with $d_i \ge Y$ form a suffix of the chain, since the chain is strictly increasing. Also $\rho(d_i)$ is nondecreasing in i. If there is no index i with $d_i \ge Y$ and $\rho(d_i) \ge 6$, there is nothing to count. Otherwise let i_0 be the first such index. Then for every $i_0 \le i < u$ we have $d_i \ge Y$ and

$$\rho(d_{i+1}) \ge \rho(d_i) \ge 6,$$

so Lemma 3.5 gives

$$\rho(d_{i+1}) \ge \rho(d_i) + 1.$$

Therefore

$$\rho(d_u) \ge \rho(d_{i_0}) + (u - i_0) \ge 6 + (u - i_0).$$

Thus the number of large terms with rank at least 6 is

$$u - i_0 + 1 \le \rho(d_u) - 5 \le L.$$

The number of large terms with $\rho(d_i) \ge 6$ is therefore at most $\rho(d_u) \le L$. The remaining large terms have $\rho(d_i) \le 5$, hence $d_i \le U_5$, hence there are $O(1)$ of them.

Putting everything together,

$$H(n) \;=\; u_{\text{small}} + u_{\text{large}} \;\le\; Y \;+\; L \;+\; O(1) \;=\; L \;+\; O(L^{1/2}) \;=\; (1 + o(1))L. \tag{33}$$

This is exactly (22). □

4. Upper bound for $h(n)$

In this section we prove:

Theorem 4.1. *For all but $o(x)$ integers $n \le x$,*

$$h(n) \;\le\; \Big(\tfrac{1}{2} + o(1)\Big) \log_* x. \tag{34}$$

The qualitative strategy is the same as for H, but the analogue of the bad-pair lemma is sharper because p, q must both be primes. The cost of "$q \equiv 1 \pmod p$ and q prime" is roughly $1/(\varphi(p)\log q)$ rather than $1/q$, and a Brun–Titchmarsh estimate makes the bad-prime-pair tail small enough to push the cutoff $\exp(\sqrt{p})$ all the way up to $Q(p) := \exp\big(\exp(p/(\log p)^2)\big)$. This faster threshold means each prime-chain step descends *two* levels of the iterated tower, halving the count.

4.1. The bad-prime-pair lemma.

For a prime $p \ge 3$ define

$$Q(p) \;:=\; \exp\Big(\exp\Big(\frac{p}{(\log p)^2}\Big)\Big). \tag{35}$$

Definition 4.2 (Bad prime pair)**.** An ordered pair of primes (p, q) is called *h-bad* if

$$p < q, \qquad q \equiv 1 \pmod p, \qquad q < Q(p). \tag{36}$$

Lemma 4.3. *Let $Y \to \infty$. The number of $n \le x$ such that some h-bad pair (p, q) with $p \ge Y$ has both $p \mid n$ and $q \mid n$ is $o(x)$.*

Proof. For fixed prime p we estimate

$$\mathcal{S}(p) \;:=\; \sum_{\substack{p<q\le Q(p)\\ q\equiv 1 \pmod p\\ q \text{ prime}}} \frac{1}{q}. \tag{37}$$

Split this into the ranges $p < q \le p^2$ and $p^2 < q \le Q(p)$.

Range $p < q \le p^2$. Here q is one of at most p primes (in fact, the trivial bound $q = p+1,\, 2p+1, \dots$ gives at most $p-1$ candidates), so

$$\sum_{\substack{p<q\le p^2\\ q\equiv 1 \pmod p\\ q \text{ prime}}} \frac{1}{q} \;\le\; \sum_{a=1}^{p-1} \frac{1}{ap+1} \;\le\; \frac{1}{p}\sum_{a=1}^{p-1}\frac{1}{a} \;\ll\; \frac{\log p}{p}. \tag{38}$$

Range $p^2 < q \le Q(p)$. By Brun–Titchmarsh (Lemma 2.4), for $t \ge 2p$,

$$\pi(t; p, 1) \;\le\; \frac{C_{BT}\, t}{\varphi(p)\log(t/p)} \;\le\; \frac{2C_{BT}\, t}{p\log(t/p)},$$

since $\varphi(p) = p - 1 \geq p/2$ for $p \geq 2$. Partial summation yields

$$\begin{aligned}
\sum_{\substack{p^2 < q \leq Q(p) \\ q \equiv 1 \pmod p \\ q \text{ prime}}} \frac{1}{q} &= \int_{p^2}^{Q(p)} \frac{d\pi(t;p,1)}{t} \\
&\leq \frac{\pi(Q(p);p,1)}{Q(p)} + \int_{p^2}^{Q(p)} \frac{\pi(t;p,1)}{t^2}\, dt \\
&\ll \frac{1}{p} \int_{p^2}^{Q(p)} \frac{dt}{t \log(t/p)} \\
&= \frac{1}{p} \Big[\log\log(t/p) \Big]_{p^2}^{Q(p)} \\
&\ll \frac{1}{p} \Big(\log\log Q(p) - \log\log p \Big). \qquad (39)
\end{aligned}$$

In the Stieltjes partial summation formula the lower boundary contribution $-\pi(p^2;p,1)/p^2$ is nonpositive and has therefore been discarded. The upper boundary term is harmless: by Brun–Titchmarsh it is

$$\frac{\pi(Q(p);p,1)}{Q(p)} \ll \frac{1}{p \log(Q(p)/p)},$$

which is absorbed by the displayed integral bound. By definition (35), $\log\log Q(p) = p/(\log p)^2$, so

$$\sum_{\substack{p^2 < q \leq Q(p) \\ q \equiv 1 \pmod p \\ q \text{ prime}}} \frac{1}{q} \ll \frac{1}{p} \cdot \frac{p}{(\log p)^2} = \frac{1}{(\log p)^2}. \qquad (40)$$

Combining (38) and (40),

$$\mathcal{S}(p) \ll \frac{\log p}{p} + \frac{1}{(\log p)^2}. \qquad (41)$$

For each h-bad pair (p,q) with $p \geq Y$, the count of $n \leq x$ with $pq \mid n$ is $\leq x/(pq)$ (recalling $\gcd(p,q) = 1$). Summing,

$$\begin{aligned}
\#\{n \leq x \colon \exists\, (p,q)\ h\text{-bad}, p \geq Y, pq \mid n\} &\leq x \sum_{p \geq Y} \frac{\mathcal{S}(p)}{p} \\
&\ll x \sum_{p \geq Y} \Big(\frac{\log p}{p^2} + \frac{1}{p(\log p)^2} \Big). \qquad (42)
\end{aligned}$$

Both series converge: $\sum_p (\log p)/p^2 < \infty$ and $\sum_p 1/(p(\log p)^2) < \infty$. Indeed, for $p \geq 3$ both prime sums are dominated by the corresponding integer sums:

$$\sum_{p \geq 3} \frac{\log p}{p^2} \leq \sum_{n \geq 3} \frac{\log n}{n^2} < \infty, \qquad \sum_{p \geq 3} \frac{1}{p(\log p)^2} \leq \sum_{n \geq 3} \frac{1}{n(\log n)^2} < \infty.$$

The second integer sum converges by the integral test. Therefore the right-hand side of (42) is $o(x)$ as $Y \to \infty$. □

4.2. **The U-tower lemma for h.** Define

$$f(t) \;:=\; C_f\,(\log\log t)\big(\log\log\log t\big)^2, \tag{43}$$

where C_f is a fixed absolute constant chosen large enough below.

Lemma 4.4. *There exists an integer $m_0^* = m_0^*(C_f) \geq 6$ such that, for all $m \geq m_0^*$,*

$$f(U_m) \;\leq\; U_{m-2}. \tag{44}$$

Proof. By (11), $\log\log U_m = T_{m-2} + \log 3$, and by (12), $\log\log\log U_m = T_{m-3} + O(T_{m-3}^{-1})$. Therefore, for $m \geq 5$,

$$\begin{aligned} f(U_m) &= C_f\,(T_{m-2} + \log 3)\big(T_{m-3} + O(T_{m-3}^{-1})\big)^2 \\ &\leq 2C_f\,T_{m-2}\,T_{m-3}^2, \end{aligned} \tag{45}$$

provided $T_{m-3} \geq 100$ (take $m \geq 6$). Now $T_{m-3} = \log T_{m-2}$, so $T_{m-3}^2 = (\log T_{m-2})^2$. We must compare this with T_{m-2}^2:

$$\frac{f(U_m)}{U_{m-2}} \;\leq\; \frac{2C_f\,T_{m-2}\,T_{m-3}^2}{T_{m-2}^3} \;=\; \frac{2C_f\,(\log T_{m-2})^2}{T_{m-2}^2}.$$

Since $(\log s)^2/s^2 \to 0$ as $s \to \infty$, this ratio tends to 0, so $f(U_m)/U_{m-2} \leq 1$ for all sufficiently large m. Concretely, choose $s_0 = s_0(C_f)$ so large that

$$\frac{2C_f(\log s)^2}{s^2} \leq 1 \qquad (s \geq s_0),$$

and then choose m_0^* so that $m_0^* \geq 6$ and $T_{m_0^*-2} \geq s_0$. This yields (44). □

Remark 4.5 (Monotonicity of the inversion function)**.** We will use that the function

$$f(t) = C_f(\log\log t)(\log\log\log t)^2$$

is increasing for all sufficiently large t, in particular for $t \geq T_3$. Indeed, writing $s = \log\log t$, we have

$$\frac{1}{C_f} f(t) = s(\log s)^2.$$

For $s > 1$,

$$\frac{d}{ds}\big(s(\log s)^2\big) = (\log s)^2 + 2\log s > 0,$$

and $s = \log\log t$ is itself increasing in t. Since $\log\log T_3 = T_1 > 1$, the asserted monotonicity holds for $t \geq T_3$.

4.3. **Proof of Theorem 4.1.** Let x be large, $L := \log_* x$, $Y := \lfloor L^{1/2} \rfloor$. By Lemma 4.3, all but $o(x)$ integers $n \leq x$ satisfy

$$\text{no } h\text{-bad pair } (p,q) \text{ with } p \geq Y \text{ has both } p \mid n \text{ and } q \mid n. \tag{$\star_h$}$$

Fix such an n, and let $p_1 < p_2 < \cdots < p_\ell$ be a prime chain witnessing $h(n) = \ell$. Let $\ell_{\text{small}} = \#\{i : p_i < Y\} \leq Y$, and $\ell_{\text{large}} = \ell - \ell_{\text{small}}$.

For each large index i with $p_i \geq Y$ and $i < \ell$, the pair (p_i, p_{i+1}) is not h-bad, so

$$p_{i+1} \;\geq\; Q(p_i) \;=\; \exp\Big(\exp\Big(\tfrac{p_i}{(\log p_i)^2}\Big)\Big). \tag{46}$$

Solving (46) for p_i:

$$\frac{p_i}{(\log p_i)^2} \;\leq\; \log\log p_{i+1}, \quad \text{so} \quad p_i \;\leq\; (\log p_i)^2 \log\log p_{i+1}.$$

We extract a clean algebraic inversion. Set $a := \log p_i$, $b := \log\log p_{i+1}$; the inequality above reads $e^a \le a^2 b$.

Since $p_i \ge Y$ and $Y \to \infty$, (46) gives

$$b = \log\log p_{i+1} \ge \frac{p_i}{(\log p_i)^2} \longrightarrow \infty,$$

uniformly for the large-index steps under consideration. Thus all estimates below that require b sufficiently large apply after increasing x if necessary.

More explicitly, after increasing x if necessary we have $Y > e^2$, and the function $t \mapsto t/(\log t)^2$ is increasing for $t > e^2$. Hence throughout the large-index part of the chain,

$$b = \log\log p_{i+1} \ge \frac{p_i}{(\log p_i)^2} \ge \frac{Y}{(\log Y)^2} \longrightarrow \infty.$$

Thus the threshold beyond which the elementary inversion below is valid may be chosen independently of the particular chain and of the index i.

We claim

$$a \;\le\; \log b \;+\; 3\log\log b \tag{47}$$

for all sufficiently large b. Indeed, the function $u \mapsto e^u/u^2$ is increasing for $u > 2$. If $a > \log b + 3\log\log b$, then

$$\frac{e^a}{a^2} \;>\; \frac{\exp(\log b + 3\log\log b)}{(\log b + 3\log\log b)^2} \;=\; b \cdot \frac{(\log b)^3}{(\log b + 3\log\log b)^2} \;>\; b$$

for all sufficiently large b, contradicting $e^a \le a^2 b$. This establishes (47). Therefore $\log p_i = a \le \log b + 3\log\log b$, and so

$$p_i \;\le\; (\log p_i)^2 \cdot b \;=\; a^2 \cdot b \;\le\; (\log b + 3\log\log b)^2 \cdot b \;\le\; C_f\, b\,(\log b)^2, \tag{48}$$

for an absolute constant $C_f > 0$ (take $C_f = 16$, say) and all sufficiently large b. Recalling $b = \log\log p_{i+1}$, we obtain

$$p_i \;\le\; C_f\,(\log\log p_{i+1})\,\big(\log\log\log p_{i+1}\big)^2 \;=\; f(p_{i+1}). \tag{49}$$

We now run the same U-rank descent as in Section 3. Let $\rho(p) := \min\{m \ge 0 : p \le U_m\}$.

Lemma 4.6 (Tower descent for h)**.** *Provided $\rho(p_{i+1}) \ge m_0^*$ (where m_0^* is from Lemma 4.4) and $p_i \ge Y \ge 100$, $i < \ell$,*

$$\rho(p_i) \;\le\; \rho(p_{i+1}) \;-\; 2. \tag{50}$$

Proof. Set $m := \rho(p_{i+1}) \ge m_0^*$. Then $p_{i+1} \le U_m$. The function $f(t) = C_f(\log\log t)(\log\log\log t)^2$ is monotone increasing for $t \ge T_3$, hence by (49),

$$p_i \;\le\; f(p_{i+1}) \;\le\; f(U_m) \;\le\; U_{m-2},$$

the last inequality being Lemma 4.4. Hence $\rho(p_i) \le m - 2$. □

Again we record the counting argument explicitly. The terms with $p_i \ge Y$ form a suffix of the prime chain, and $\rho(p_i)$ is nondecreasing in i. If there is no index i with $p_i \ge Y$ and $\rho(p_i) \ge m_0^*$, then there are no large high-rank terms to count. Otherwise let i_0 be the first such index. For every $i_0 \le i < \ell$ we have $p_i \ge Y$ and

$$\rho(p_{i+1}) \ge \rho(p_i) \ge m_0^*,$$

so Lemma 4.6 gives

$$\rho(p_{i+1}) \ge \rho(p_i) + 2.$$

Consequently

$$\rho(p_\ell) \ge \rho(p_{i_0}) + 2(\ell - i_0) \ge m_0^* + 2(\ell - i_0).$$

Thus the number of large terms with rank at least m_0^* is at most

$$\ell - i_0 + 1 \le 1 + \frac{\rho(p_\ell) - m_0^*}{2} \le \frac{L}{2} + O(1).$$

Since $p_\ell \le n \le x \le T_L \le U_L$, we have $\rho(p_\ell) \le L$. The number of large terms i with $\rho(p_i) \ge m_0^*$ is therefore at most $L/2 + O(1)$, since each such large term is at least 2 levels below the next. The remaining large terms have $\rho(p_i) < m_0^*$, hence $p_i < U_{m_0^*-1} = O(1)$, so there are $O(1)$ of them.

Adding $\ell_{\text{small}} \le Y = \lfloor L^{1/2} \rfloor$,

$$h(n) \;=\; \ell \;\le\; \tfrac{L}{2} + O(1) + Y \;=\; \tfrac{L}{2} + O(L^{1/2}) \;=\; \big(\tfrac{1}{2} + o(1)\big)L, \tag{51}$$

proving (34). □

5. Lower bound for $h(n)$

We now prove the matching lower bound:

Theorem 5.1. *For all but $o(x)$ integers $n \le x$,*

$$h(n) \;\ge\; \Big(\tfrac{1}{2} - o(1)\Big) \log_* x. \tag{52}$$

The strategy is to construct, for "random" n, a long prime chain $p_1 < \cdots < p_R$ greedily: at each stage we use Siegel–Walfisz to ensure that the next prime in the residue class $1 \bmod p_j$, lying inside a controlled tower window, exists with overwhelming probability. We then transfer to integers via the CRT.

Throughout this section we put

$$L := \log_* x.$$

All choices of scales below are functions of this parameter L, and hence of x.

5.1. The prime-successor mass lemma.

Lemma 5.2 (Uniform prime-successor mass)**.** *There are absolute constants $c_0 > 0$, $y_0 \ge 1$ such that the following holds. For every $y \ge y_0$, every prime $p \le y$, and every $\eta \ge 2$,*

$$\sum_{\substack{\exp(\exp y) < q \le \exp(\exp(\eta y)) \\ q \equiv 1 \pmod p \\ q \text{ prime}}} \frac{1}{q} \;=\; \frac{(\eta - 1)y}{\varphi(p)} \;+\; O\big(\exp(-c_0 \exp(y/2))\big),$$

uniformly in p, y, η. In particular,

$$\sum_{\substack{\exp(\exp y) < q \le \exp(\exp(\eta y)) \\ q \equiv 1 \pmod p \\ q \text{ prime}}} \frac{1}{q} \;\ge\; \frac{1}{2}\,\frac{(\eta - 1)y}{p}$$

for all sufficiently large y.

Proof. Let

$$Z = \exp(\exp y), \qquad X = \exp(\exp(\eta y)).$$

For $t \ge Z$, $\log t \ge \exp y$, hence $p \le y \le (\log Z)$ for all sufficiently large y, so Lemma 2.1 with $A_0 = 1$ applies on $[Z, X]$. By (14),

$$\sum_{\substack{Z < q \le X \\ q \equiv 1 \pmod p}} \frac{1}{q} \;=\; \frac{1}{\varphi(p)} \int_Z^X \frac{dt}{t \log t} \;+\; O\Big(\exp(-c_0 \sqrt{\log Z})\Big).$$

The integral evaluates to $\log\log X - \log\log Z = (\eta-1)y$, and $\sqrt{\log Z} = \exp(y/2)$, so the error is $O(\exp(-c_0\exp(y/2)))$. This proves the asymptotic formula.

Since $p \le y$, $\varphi(p) = p-1 \le p$, and $(\eta-1)y/p \ge 1$ for $\eta \ge 2$ and $p \le y$. For sufficiently large y, the error term is therefore smaller than half the main lower bound, giving

$$\sum_{\substack{Z<q\le X\\ q\equiv 1 \pmod p}} \frac{1}{q} \;\ge\; \frac{1}{2}\,\frac{(\eta-1)y}{p}. \qquad \square$$

5.2. **Tower scale for h and the greedy construction.** Fix $B := L^2$ and the integers

$$m_0 \;=\; \lfloor L^{1/2}\rfloor, \qquad m_j \;=\; m_0 + 2(j-1) \quad (j \ge 1), \tag{53}$$

so that m_j increments by 2. Let R be the largest integer with $m_R \le L-2$; then

$$R \;=\; \frac{L-m_0-2}{2} + O(1) \;=\; \Big(\tfrac{1}{2} - o(1)\Big)L. \tag{54}$$

Define

$$y_j \;=\; T_{m_j}/B \qquad (1 \le j \le R+1). \tag{55}$$

Since $T_{m_{j+1}} = T_{m_j+2} = \exp\big(\exp(T_{m_j})\big) = \exp(\exp(By_j))$, we have

$$y_{j+1} \;=\; \frac{T_{m_{j+1}}}{B} \;=\; \frac{\exp(\exp(By_j))}{B}. \tag{56}$$

In particular,

$$\log\log y_{j+1} \;=\; \log\big(\exp(By_j) - \log B\big) \;=\; By_j + O(1). \tag{57}$$

Lemma 5.3. *Work in the independent prime model on primes $r \le y_R$. The probability that the following event $\mathcal{C}_h$* fails *is $o(1)$:*

> $\mathcal{C}_h$: *there exist primes $p_1 < \cdots < p_R$ all selected, with $p_j \le y_j$ and $p_{j+1} \equiv 1 \pmod{p_j}$ for $1 \le j < R$.*

Proof. We expose primes in stages and make deterministic choices throughout. If there is at least one selected prime $p \le y_1$, define p_1 to be the least selected prime $\le y_1$; otherwise the construction fails at the initial stage.

The probability of initial failure is

$$\prod_{p\le y_1}(1-1/p) \;\le\; \exp\Big(-\sum_{p\le y_1} 1/p\Big) \;=\; \exp(-\log\log y_1 + O(1)). \tag{58}$$

Since $y_1 = T_{m_0}/B$ and $m_0 \to \infty$, we have $\log\log y_1 = T_{m_0-2} + O(\log B) \to \infty$, so (58) is $o(1)$.

Now suppose that $p_1, \ldots, p_j$ have already been defined, with $p_i \le y_i$ and $p_{i+1} \equiv 1 \pmod{p_i}$ for $1 \le i < j$. Define the candidate set

$$\mathcal{Q}_j(p_j) \;:=\; \big\{q \text{ prime} : q \equiv 1 \pmod{p_j},\ \exp(\exp y_j) < q \le \exp(\exp((B/2)y_j))\big\}.$$

If at least one prime in $\mathcal{Q}_j(p_j)$ is selected, define p_{j+1} to be the least selected prime in $\mathcal{Q}_j(p_j)$; otherwise the construction fails at stage j.

By induction, every prime whose selection has been inspected before stage j is at most y_j: the initial step inspects only primes $\le y_1$, and the search interval at stage $r < j$ is contained in $(\exp(\exp y_r), y_{r+1}]$, with $y_{r+1} \le y_j$. On the other hand every prime in $\mathcal{Q}_j(p_j)$ is $> \exp(\exp y_j) > y_j$. Thus the primes inspected at stage j are disjoint from all previously inspected primes. This justifies conditioning on the entire past while retaining independent Bernoulli selection for the primes in $\mathcal{Q}_j(p_j)$.

For L large, this search interval is contained in $(\exp(\exp y_j),\, y_{j+1}]$, since by (56)

$$\exp(\exp((B/2)y_j)) \;\ll\; \frac{\exp(\exp(By_j))}{B} \;=\; y_{j+1}.$$

Hence, on success at stage j, $p_{j+1} \le y_{j+1}$. Also $p_{j+1} > \exp(\exp y_j) > y_j \ge p_j$, so the chain remains strictly increasing.

Apply Lemma 5.2 with $y = y_j$, $\eta = B/2$, and modulus $p = p_j \le y_j$. Since $B = L^2$, we have $\eta \ge 2$ for large L, so

$$\sum_{q\in\mathcal{Q}_j(p_j)} \frac{1}{q} \;\ge\; \frac{1}{2}\,\frac{(B/2-1)y_j}{p_j} \;\ge\; \frac{B}{8}, \tag{59}$$

for $B \ge 4$, since $p_j \le y_j$.

The construction up to stage j depends only on selected primes lying below the present search interval, whereas every prime in $\mathcal{Q}_j(p_j)$ is $> \exp(\exp y_j) > y_j \ge p_j$. Hence, conditional on the past, the selection events for primes in $\mathcal{Q}_j(p_j)$ are still independent Bernoulli events with probabilities $1/q$. Therefore

$$\mathbb{P}(\text{stage } j \text{ fails} \mid \text{past}) \;\le\; \prod_{q\in\mathcal{Q}_j(p_j)} \big(1-1/q\big) \;\le\; \exp\Big(-\sum_{q\in\mathcal{Q}_j(p_j)} 1/q\Big) \;\le\; \exp(-B/8) \;=\; \exp(-L^2/8).$$

Union-bounding over $1 \le j < R \le L$,

$$\mathbb{P}(\mathcal{C}_h^c) \;\le\; o(1) + L\cdot\exp(-L^2/8) \;=\; o(1). \tag{60}$$

□

5.3. **Proof of Theorem 5.1.** We apply Lemma 2.7 with the cutoff $P := y_R$. The event $\mathcal{C}_h$ depends only on which primes $p \le y_R$ divide n, so it is a union of residue classes mod $M := \prod_{p\le y_R} p$. We need $M = o(x)$.

Indeed, the initial choice of p_1 only inspects primes $p \le y_1 \le y_R$. At stage $j < R$, the search interval $\mathcal{Q}_j(p_j)$ is contained in $(\exp(\exp y_j), y_{j+1}]$, as shown above, and hence contains only primes $\le y_R$. The deterministic rule "choose the least selected prime" makes the event $\mathcal{C}_h$ a function only of the indicator vector $\big(\mathbf{1}_{p|n}\big)_{p\le y_R}$. No higher prime powers are used.

Since $m_R \le L-2$, $y_R \le T_{L-2}/B$. On the other hand, by the definition of $L = \log_* x$ we have

$$T_{L-1} < x \le T_L$$

for all sufficiently large L. Taking one logarithm in the lower inequality and using $\log T_{L-1} = T_{L-2}$ gives

$$\log x \;>\; T_{L-2}. \tag{61}$$

Hence $y_R \le T_{L-2}/L^2 = o(\log x)$. By Chebyshev (Lemma 2.6), $\log M = \sum_{p\le y_R} \log p \le C_\theta y_R = o(\log x)$, so $M = x^{o(1)} = o(x)$.

We also record explicitly that the constructed prime divisibilities are compatible with the restriction $n \le x$. If $\mathcal{C}_h$ is realized by an integer n, then each constructed prime p_j is one of the primes whose indicator is 1, and hence $p_j \mid n$. Since the primes $p_1, \dots, p_R$ are distinct and all lie below $P = y_R$, their product is at most

$$\prod_{p\le y_R} p = M = o(x).$$

Thus the CRT transfer is prescribing a squarefree divisor of size $< x$ for all sufficiently large x; equivalently, by Remark 2.8, no extra size obstruction is present.

By Lemma 2.7, the density of $n \le x$ for which $\mathcal{C}_h$ holds is $\mathbb{P}_{\text{prod}}(\mathcal{C}_h) + O(M/x) = 1 - o(1)$. Whenever $\mathcal{C}_h$ holds, the chain witnesses $h(n) \ge R = (\frac{1}{2} - o(1))L$. This proves (52). □

6. The subset-product successor lemma

This section contains the key probabilistic ingredient for the lower bound on $H(n)$. We work in two steps. First, an abstract statement about random elements of finite abelian groups (Lemma 6.1). Then a number-theoretic application that produces composite-divisor successors (Lemma 6.5).

6.1. Abstract subset-product lemma.

Let G be a finite abelian group, written multiplicatively, with identity 1_G and order $N = |G|$.

In the next lemma no generation hypothesis is imposed on the sampled elements: they are only required to be independent and uniform. In the number-theoretic application the residue classes coming from good blocks need not generate G.

Lemma 6.1. *Let $g_1, \dots, g_K$ be independent uniformly distributed elements of G. Define*

$$Z \;:=\; \#\Big\{\emptyset \neq S \subseteq \{1,\dots,K\} : \prod_{i\in S} g_i = 1_G\Big\}. \tag{62}$$

Then

$$\mathbb{E}[Z] \;=\; \frac{2^K-1}{N}, \tag{63}$$

$$\mathrm{Var}[Z] \;\le\; \mathbb{E}[Z], \tag{64}$$

$$\mathbb{P}(Z=0) \;\le\; \frac{N}{2^K-1}. \tag{65}$$

Proof. For each nonempty $S \subseteq [K]$, the random variable $X_S := \prod_{i\in S} g_i$ is uniformly distributed on G (by independence of the g_i and translation invariance), so $\mathbb{P}(X_S = 1_G) = 1/N$. Summing,

$$\mathbb{E}[Z] \;=\; \sum_{\emptyset\neq S\subseteq[K]} \mathbb{P}(X_S = 1_G) \;=\; \frac{2^K-1}{N},$$

proving (63).

To bound the variance we show that for any two distinct nonempty subsets $S \neq T$ of $[K]$, the random variables X_S and X_T are *independent*. Pick

$$i_0 \;\in\; S \triangle T, \tag{66}$$

the symmetric difference (which is nonempty since $S \neq T$). Without loss of generality $i_0 \in S \setminus T$ (else swap S, T). Condition on the joint value of $\{g_j : j \neq i_0\}$. Under this conditioning X_T is determined (because $i_0 \notin T$), so X_T takes a fixed value; meanwhile $X_S = g_{i_0} \cdot \prod_{i\in S\setminus\{i_0\}} g_i$ depends on g_{i_0}, which under our conditioning is uniformly distributed on G. Thus $X_S \mid \{g_j : j \neq i_0\}$ is uniform on G, hence in particular conditionally independent of X_T. Marginalizing,

$$X_S \text{ and } X_T \text{ are independent, both uniform on } G. \tag{67}$$

Equivalently, for any $x_0, y_0 \in G$, conditioning on all g_j with $j \neq i_0$ gives

$$\begin{aligned}\mathbb{P}(X_S = x_0,\; X_T = y_0) &= \mathbb{E}\big[\mathbf{1}_{\{X_T = y_0\}}\, \mathbb{P}(X_S = x_0 \mid \{g_j : j \neq i_0\})\big] \\ &= \frac{1}{N}\, \mathbb{P}(X_T = y_0) = \frac{1}{N^2}.\end{aligned}$$

This gives the asserted independence for every distinct pair $S \neq T$. Therefore $\mathbb{P}(X_S = X_T = 1_G) = 1/N^2$. Hence

$$\mathbb{E}[Z^2] \;=\; \sum_S \mathbb{P}(X_S = 1) + \sum_{S\neq T} \mathbb{P}(X_S = X_T = 1) \;=\; \frac{2^K-1}{N} + \frac{(2^K-1)(2^K-2)}{N^2}.$$

Subtracting $(\mathbb{E}Z)^2 = (2^K-1)^2/N^2$,

$$\operatorname{Var} Z \;=\; \frac{2^K-1}{N} - \frac{2^K-1}{N^2} \;=\; \frac{(2^K-1)(N-1)}{N^2} \;\le\; \frac{2^K-1}{N} \;=\; \mathbb{E}Z, \tag{68}$$

proving (64).

Finally, by Chebyshev's inequality,

$$\mathbb{P}(Z=0) \;\le\; \mathbb{P}\big(|Z-\mathbb{E}Z| \ge \mathbb{E}Z\big) \;\le\; \frac{\operatorname{Var} Z}{(\mathbb{E}Z)^2} \;\le\; \frac{1}{\mathbb{E}Z} \;=\; \frac{N}{2^K-1},$$

proving (65). □

Remark 6.2 (Near-uniform version)**.** For our application, the g_i are not exactly uniform on G but only close to uniform in total variation. We record the precise reduction. Let u_G denote the uniform probability measure on G, and suppose that the random variables $g_1,\dots,g_K$ are independent, with laws μ_i satisfying

$$d_{\mathrm{TV}}(\mu_i, u_G) \le \varepsilon \qquad (1 \le i \le K).$$

For each i, choose a maximal coupling of g_i with a uniform random variable $\widetilde{g}_i \sim u_G$ such that

$$\mathbb{P}(g_i \ne \widetilde{g}_i) = d_{\mathrm{TV}}(\mu_i, u_G) \le \varepsilon,$$

and choose these couplings independently for different i. Then $\widetilde{g}_1,\dots,\widetilde{g}_K$ are independent and exactly uniform on G, and by the union bound

$$\mathbb{P}\big((g_1,\dots,g_K) \ne (\widetilde{g}_1,\dots,\widetilde{g}_K)\big) \le K\varepsilon.$$

Let Z be the number of identity subset-products formed from the g_i, and let $\widetilde{Z}$ be the corresponding number for the $\widetilde{g}_i$. The events $\{Z=0\}$ and $\{\widetilde{Z}=0\}$ can differ only if at least one g_i differs from $\widetilde{g}_i$. Hence Lemma 6.1 gives

$$\mathbb{P}(Z=0) \le \mathbb{P}(\widetilde{Z}=0) + K\varepsilon \le \frac{N}{2^K-1} + K\varepsilon. \tag{69}$$

More generally, if the ith total-variation gap is ε_i, the additive error is $\sum_{i=1}^K \varepsilon_i$.

6.2. **The composite-successor lemma.** In the next lemma the subscript A in $O_A(\cdot)$ refers only to the fixed absolute window parameter used later in Lemma 6.5. In the application below we take $A=20$, so all such constants are absolute.

Lemma 6.3 (Weighted residue equidistribution in a block)**.** *Let $d \le y$, let $G := (\mathbb{Z}/d\mathbb{Z})^\times$, and let $\mathcal{B} = (\alpha,\beta]$ be one of the blocks constructed below in the proof of Lemma 6.5 (so $\alpha \ge \exp y$ and the reciprocal-prime mass satisfies $\mu(\mathcal{B}) := \sum_{q\in\mathcal{B}} 1/q = 1 + O(1/y)$). For every reduced residue class $a \bmod d$,*

$$\sum_{\substack{q\in\mathcal{B}\\ q\equiv a \pmod d}} \frac{1}{q} \;=\; \frac{\mu(\mathcal{B})}{\varphi(d)} \;+\; O_A\big(\exp(-c\sqrt{y})\big),$$

uniformly in $d \le y$, $a \in G$, and the block $\mathcal{B}$. The same estimate holds with $1/q$ replaced by $1/(q-1)$. Consequently, conditional on the event that exactly one prime in $\mathcal{B}$ is selected, the residue class of that prime modulo d has total variation distance $O(y^{-1/2})$ from the uniform distribution on G.

Proof. The case $d=1$ is trivial, since G has one element and the total-variation distance to the uniform distribution is zero. We may assume $d \ge 2$. Since every prime in $\mathcal{B}$ is $> \exp y > d$, each such prime is coprime to d and therefore defines an element of G.

Since $d \le y \le \log \alpha$, Lemma 2.1 with $A_0 = 1$ applies uniformly on $(\alpha, \beta]$. By (14),

$$\sum_{\substack{\alpha < q \le \beta \\ q \equiv a \pmod d}} \frac{1}{q} \;=\; \frac{1}{\varphi(d)} \int_\alpha^\beta \frac{dt}{t \log t} \;+\; O\big(\exp(-c_0\sqrt{\log \alpha})\big).$$

Since $\log \alpha \ge y$, the error is $O(\exp(-c_0\sqrt{y}))$. Write

$$I_{\mathcal{B}} := \int_\alpha^\beta \frac{dt}{t \log t}.$$

For each reduced residue class $a \bmod d$ we have

$$\sum_{\substack{q \in \mathcal{B} \\ q \equiv a \pmod d}} \frac{1}{q} = \frac{I_{\mathcal{B}}}{\varphi(d)} + O\big(\exp(-c_0\sqrt{y})\big),$$

uniformly in $d \le y$, a, and $\mathcal{B}$. Summing this estimate over all $a \in (\mathbb{Z}/d\mathbb{Z})^\times$ gives

$$\mu(\mathcal{B}) = I_{\mathcal{B}} + O\big(\varphi(d)\exp(-c_0\sqrt{y})\big).$$

Dividing by $\varphi(d)$ and substituting back yields

$$\frac{I_{\mathcal{B}}}{\varphi(d)} = \frac{\mu(\mathcal{B})}{\varphi(d)} + O\big(\exp(-c_0\sqrt{y})\big),$$

and hence

$$\sum_{\substack{q \in \mathcal{B} \\ q \equiv a \pmod d}} \frac{1}{q} = \frac{\mu(\mathcal{B})}{\varphi(d)} + O\big(\exp(-c_0\sqrt{y})\big).$$

After decreasing c_0 if necessary, this is the stated $O_A(\exp(-c\sqrt{y}))$ estimate.

For the weight $1/(q-1)$, note that

$$\sum_{q \in \mathcal{B}} \left| \frac{1}{q-1} - \frac{1}{q} \right| \;\ll\; \sum_{q > \exp y} \frac{1}{q^2} \;\ll\; \exp(-y),$$

and the same bound holds after restricting to any residue class.

Now condition on the event $\mathcal{G}_{\mathcal{B}}$ that exactly one prime in $\mathcal{B}$ is selected, and let $Q_{\mathcal{B}}$ denote that unique selected prime. For $a \in G$, we have

$$\begin{aligned}
\mathbb{P}(Q_{\mathcal{B}} \equiv a \pmod d \mid \mathcal{G}_{\mathcal{B}}) &= \frac{\sum_{\substack{q \in \mathcal{B} \\ q \equiv a \pmod d}} \frac{1}{q} \prod_{r \in \mathcal{B},\ r \ne q} (1 - 1/r)}{\sum_{q \in \mathcal{B}} \frac{1}{q} \prod_{r \in \mathcal{B},\ r \ne q} (1 - 1/r)} \\
&= \frac{\sum_{\substack{q \in \mathcal{B} \\ q \equiv a \pmod d}} \frac{1}{q-1}}{\sum_{q \in \mathcal{B}} \frac{1}{q-1}},
\end{aligned}$$

because the common factor $\prod_{r \in \mathcal{B}} (1 - 1/r)$ cancels.

By the estimate just proved with the weight $1/(q-1)$, the numerator is

$$\frac{\mu(\mathcal{B})}{\varphi(d)} + O_A\big(\exp(-c\sqrt{y})\big),$$

while the denominator is

$$\mu(\mathcal{B}) + O(\exp(-y)) = 1 + O(1/y).$$

Since $\mu(\mathcal{B}) = 1 + O(1/y)$, it follows that

$$\mathbb{P}(Q_{\mathcal{B}} \equiv a \pmod{d} \mid \mathcal{G}_{\mathcal{B}}) = \frac{1}{\varphi(d)} + O\left(\frac{1}{y\varphi(d)}\right) + O_A\big(\exp(-c\sqrt{y})\big).$$

Consequently the total-variation distance from the uniform distribution on G is at most

$$\frac{1}{2}\sum_{a\in G}\left|\mathbb{P}(Q_{\mathcal{B}} \equiv a \pmod{d} \mid \mathcal{G}_{\mathcal{B}}) - \frac{1}{\varphi(d)}\right| \ll \frac{1}{y} + y\exp(-c\sqrt{y}) \ll y^{-1/2},$$

for all sufficiently large y. □ □

Remark 6.4 (Uniformity under conditioning on good blocks)**.** The total-variation estimate in Lemma 6.3 is uniform in $d \le y$, in the reduced residue class, and in the block $\mathcal{B}$ among the blocks constructed in Lemma 6.5. Consequently, if we condition on any fixed collection of block indices being good, the unique selected primes in those blocks remain independent across the chosen blocks, and each of their residue classes modulo d is still $O(y^{-1/2})$-close to uniform on $(\mathbb{Z}/d\mathbb{Z})^\times$. This is because the blocks are disjoint, and inside a chosen block the conditioning is only the event that exactly one prime of that block is selected.

The same conclusion holds when the block indices are chosen as the first K good blocks. Indeed, conditioning on a realized value $k_1 < \cdots < k_K$ of these indices imposes only the events that $\mathcal{B}_{k_i}$ is good and that certain disjoint intervening blocks are not good. The latter events involve no primes from the blocks $\mathcal{B}_{k_i}$, and within each $\mathcal{B}_{k_i}$ the only conditioning is that exactly one prime is selected. Therefore the unique selected primes from the realized good blocks remain independent, and their residue laws retain the same uniform $O(y^{-1/2})$ total-variation bound.

Lemma 6.5. *There exist absolute constants $A > 10$, $c > 0$, and $y_0 \ge 2$ with the following property. Let $y \ge y_0$ and $1 \le d \le y$. Work in the independent prime model in which each prime q is selected independently with probability $1/q$. Then, with probability at least $1 - O(y^{-c})$, uniformly in $d \le y$, there exists a squarefree product e of* selected *primes lying entirely in $\big(\exp y, \exp(y^{A-1})\big]$, such that*

$$e \equiv 1 \pmod{d}, \qquad e > d, \qquad e \le \exp(y^A). \tag{70}$$

Proof. Set $A := 20$ (any sufficiently large constant works; the value 20 comfortably satisfies all numerical thresholds below), and take y_0 large enough for the estimates below.

We first dispose of the case $d = 1$. In this case the congruence condition $e \equiv 1 \pmod{d}$ is vacuous. By Mertens' theorem,

$$\sum_{\exp y < q \le \exp(y^2)} \frac{1}{q} = \log\log(\exp(y^2)) - \log\log(\exp y) + O(1/y) = \log y + O(1/y).$$

Hence the probability that no prime in $(\exp y, \exp(y^2)]$ is selected is at most

$$\exp\left(-\sum_{\exp y < q \le \exp(y^2)} \frac{1}{q}\right) = \exp\big(-\log y + O(1/y)\big) \ll y^{-1}.$$

On the complementary event, any selected prime $q \in (\exp y, \exp(y^2)]$ gives $e = q$, which is squarefree, satisfies $e > 1 = d$, satisfies $e \equiv 1 \pmod{1}$, lies in $\big(\exp y, \exp(y^{A-1})\big]$, and obeys $e \le \exp(y^2) \le \exp(y^A)$. Thus the lemma holds for $d = 1$, after decreasing c if necessary.

We may therefore assume $d \ge 2$. Write $G := (\mathbb{Z}/d\mathbb{Z})^\times$ and $N := \varphi(d) \le d \le y$.

Step 1: Mertens reciprocal mass on the prime window. By Lemma 2.5,

$$\sum_{\exp y<q\le\exp(y^{A-1})} \frac{1}{q} \;=\; \log\log\bigl(\exp(y^{A-1})\bigr) - \log\log(\exp y) + O(1) \;=\; (A-2)\log y + O(1). \tag{71}$$

Step 2: Block decomposition. Recall $A = 20$. Fix $\gamma = 15$ and set

$$M \;:=\; \lfloor \gamma \log y \rfloor. \tag{72}$$

Define real endpoints α_k by

$$\log\log\alpha_k \;=\; \log y + k - 1, \qquad 1 \le k \le M+1.$$

Thus $\alpha_1 = \exp y$, and

$$\alpha_{M+1} \;=\; \exp\bigl(y\, e^{M}\bigr) \;\le\; \exp\bigl(y^{\gamma+1}\bigr) \;\le\; \exp(y^{A-1})$$

for all large y, since $\gamma + 1 = 16 < A - 1 = 19$. Let

$$\mathcal{B}_k \;:=\; \bigl(\alpha_k, \alpha_{k+1}\bigr]$$

denote the set of primes in this interval. By Mertens' theorem (Lemma 2.5),

$$\mu_k \;:=\; \sum_{q\in\mathcal{B}_k} \frac{1}{q} \;=\; \log\log\alpha_{k+1} - \log\log\alpha_k + O(1/\log\alpha_k) \;=\; 1 + O(1/y), \tag{73}$$

uniformly in k. The total mass absorbed by the first M blocks is $M\cdot(1+O(1/y)) = \gamma\log y\cdot(1+o(1))$, which fits inside the window since $\gamma < A-2$.

Step 3: Good blocks. Call a block $\mathcal{B}_k$ *good* if exactly one prime in $\mathcal{B}_k$ is selected. Put

$$\mu_k := \sum_{q\in\mathcal{B}_k}\frac{1}{q} = 1 + O(1/y), \qquad \Pi_k := \prod_{q\in\mathcal{B}_k}\Bigl(1-\frac{1}{q}\Bigr).$$

Then

$$\begin{aligned}\mathbb{P}(\mathcal{B}_k \text{ is good}) &= \sum_{q\in\mathcal{B}_k}\frac{1}{q}\prod_{q'\in\mathcal{B}_k\setminus\{q\}}\Bigl(1-\frac{1}{q'}\Bigr)\\ &= \Pi_k\sum_{q\in\mathcal{B}_k}\frac{1/q}{1-1/q} = \Pi_k\sum_{q\in\mathcal{B}_k}\frac{1}{q-1}.\end{aligned}$$

Moreover,

$$\sum_{q\in\mathcal{B}_k}\frac{1}{q-1} = \sum_{q\in\mathcal{B}_k}\frac{1}{q} + O\Bigl(\sum_{q>\exp y}\frac{1}{q^2}\Bigr) = \mu_k + O(e^{-y}) = 1 + O(1/y),$$

and

$$\log\Pi_k = \sum_{q\in\mathcal{B}_k}\log\Bigl(1-\frac{1}{q}\Bigr) = -\mu_k + O\Bigl(\sum_{q>\exp y}\frac{1}{q^2}\Bigr) = -1 + O(1/y).$$

Therefore

$$\mathbb{P}(\mathcal{B}_k \text{ is good}) = e^{-1} + O(1/y).$$

Hence, writing $\rho := e^{-1}$,

$$\mathbb{P}(\mathcal{B}_k \text{ is good}) \;=\; \rho \;+\; O(1/y) \qquad \text{(uniformly in } k). \tag{74}$$

The events $\{\mathcal{B}_k \text{ is good}\}$ for different k depend on *disjoint* sets of primes, so they are mutually independent.

Step 4: Chernoff for the number of good blocks. Let I_k be the indicator of the event that $\mathcal{B}_k$ is good, and let

$$W := \sum_{k=1}^{M} I_k.$$

The variables I_k are independent, since the blocks use disjoint sets of primes. By (74),

$$\mathbb{E}W = \sum_{k=1}^{M} \mathbb{P}(\mathcal{B}_k \text{ is good}) = \rho M + O(M/y) = \rho\gamma \log y + O(1).$$

Choose

$$\kappa \ := \ \tfrac{\rho\gamma}{2} > \frac{1}{\log 2} + 1, \qquad \text{i.e. } \gamma > 2(\rho^{-1})\big(\tfrac{1}{\log 2} + 1\big). \tag{75}$$

Numerically $\rho = e^{-1} \approx 0.368$, $1/\log 2 \approx 1.443$, so it suffices to take $\gamma \geq 14$. We also require $\gamma < A - 2 = 18$, so $\gamma \in [14, 18)$ is admissible; we keep the choice $\gamma = 15$.

For all sufficiently large y, the threshold $\kappa \log y$ is at most $(2/3)\mathbb{E}W$. The Chernoff bound for sums of independent Bernoulli variables therefore gives

$$\mathbb{P}\big(W \ < \ \kappa \log y\big) \leq \mathbb{P}\big(W < (2/3)\mathbb{E}W\big) \leq \exp(-\mathbb{E}W/18) = O(y^{-c_1}) \tag{76}$$

for some constant $c_1 = c_1(\gamma) > 0$. The use of Chernoff here is justified by Lemma 2.9: the indicators I_k are independent because the prime blocks are disjoint, and they need not be identically distributed. The estimate uses only their independence and the value of $\mathbb{E}W$.

Step 5: Conditional residue distribution. Put

$$K := \lceil \kappa \log y \rceil.$$

On the event $W \geq K$, let

$$k_1 < k_2 < \cdots < k_K$$

be the first K indices for which $\mathcal{B}_{k_i}$ is good. From each good block $\mathcal{B}_{k_i}$ we extract the unique selected prime q_{k_i}, which lies in $\mathcal{B}_{k_i} \subset (\exp y, \exp(y^{A-1})]$, hence $q_{k_i} \bmod d \in G$.

The important point is that the random indices $k_1, \ldots, k_K$ are determined only by the good/block indicators, i.e. by the events "exactly one prime is selected in $\mathcal{B}_k$". They are not determined by the residue classes of the unique selected primes. More formally, condition on any possible value of the index vector $(k_1, \ldots, k_K)$. Since the blocks are disjoint, the selected prime in each conditioned-good block is independent of the selected primes in the other conditioned-good blocks, and the law inside each such block is the law of the unique selected prime conditional on that block being good.

Equivalently, if $C_1, \ldots, C_K$ are arbitrary subsets of G, then on conditioning on the event that the first K good block indices are precisely $k_1 < \cdots < k_K$, independence of the disjoint prime blocks gives

$$\mathbb{P}\big(q_{k_i} \bmod d \in C_i \ (1 \leq i \leq K) \mid k_1, \ldots, k_K\big) = \prod_{i=1}^{K} \mathbb{P}\big(q_{k_i} \bmod d \in C_i \mid \mathcal{B}_{k_i} \text{ is good}\big).$$

The information that the intervening blocks are not good concerns only those intervening blocks and therefore does not affect the conditional distribution inside the good blocks $\mathcal{B}_{k_i}$.

By Lemma 6.3, for each i this conditional residue distribution is $O(y^{-1/2})$-close in total variation to the uniform distribution on $G = (\mathbb{Z}/d\mathbb{Z})^\times$, uniformly in the realized index k_i. Thus, conditional on the realized index vector, the residues

$$g_i := q_{k_i} \bmod d \qquad (1 \leq i \leq K)$$

are independent and each is ε_y-close to uniform on G, where

$$\varepsilon_y := O(y^{-1/2}).$$

This conclusion is uniform over the conditioned index vector $(k_1, \dots, k_K)$ by Remark 6.4.

Step 6: Apply the subset-product lemma. The following estimate is first taken conditional on the event $W \geq K$ and on an arbitrary realized index vector $k_1 < \cdots < k_K$ for the first K good blocks. By the preceding step and Remark 6.4, under this conditioning the residues

$$g_i := q_{k_i} \bmod d \qquad (1 \leq i \leq K)$$

are independent and each has total-variation distance $O(y^{-1/2})$ from the uniform distribution on G. The bound obtained below is uniform in the realized index vector, so it remains valid after averaging over the possible realized vectors.

With Lemma 6.1 (and Remark 6.2) applied to the K residues $g_i := q_{k_i} \bmod d \in G$,

$$\begin{aligned}
\mathbb{P}\big(\nexists\, \emptyset \neq S \subseteq [K] : \textstyle\prod_{i\in S} g_i = 1_G\big) &\leq \frac{N}{2^K - 1} + K\varepsilon_y \\
&\leq \frac{y}{2^{\kappa \log y - 1}} + O\big((\log y) y^{-1/2}\big) \\
&= y^{1-\kappa\log 2} \cdot 2 + O\big(y^{-1/2}(\log y)\big) \\
&= O(y^{-c_2}),
\end{aligned} \tag{77}$$

The estimate above is conditional on the event $W \geq K$ and on any fixed realization of the first K good block indices. Since the bound is uniform in that realization, averaging over the possible index vectors preserves the same bound. where one may take

$$c_2 = \min\big(\kappa \log 2 - 1,\ \tfrac{1}{3}\big) > 0$$

by (75), since $(\log y) y^{-1/2} = O(y^{-1/3})$.

Step 7: Putting it together. Combining Steps 4 and 6,

$$\mathbb{P}(\text{no such product } e \text{ exists}) \ \leq\ O(y^{-c_1}) + O(y^{-c_2}) \ =\ O(y^{-c}), \qquad c := \min(c_1, c_2) > 0. \tag{78}$$

When the construction succeeds, the resulting product $e = \prod_{i\in S} q_{k_i}$ uses at most $K = \lceil \kappa \log y \rceil$ primes. Since $\kappa = \rho\gamma/2 < \gamma$ and y is large, we have $K \leq \gamma \log y$. Each prime factor is at most $\exp(y^{A-1})$, so

$$\log e \ \leq\ \sum_{i\in S} \log q_{k_i} \ \leq\ K \cdot y^{A-1} \ \leq\ \gamma(\log y) \cdot y^{A-1} \ \leq\ y^A, \tag{79}$$

since $(\log y) \leq y$ for $y \geq 1$. Also $e > \exp y > y \geq d$ since $S \neq \emptyset$ and each prime in the product exceeds $\exp y$. The product is squarefree because the primes q_{k_i} in distinct good blocks are distinct (the blocks are disjoint). Finally $e \equiv 1 \pmod d$ by the subset-product condition. Hence e satisfies (70). □ □

7. Lower bound for $H(n)$

We now prove the missing matching lower bound:

Theorem 7.1. *For all but $o(x)$ integers $n \leq x$,*

$$H(n) \ \geq\ \big(1 - o(1)\big) \log_* x. \tag{80}$$

7.1. **Tower scale for H.** Throughout this section we again write

$$L := \log_* x.$$

Let $A > 10$ be the constant from Lemma 6.5, and choose

$$B \;:=\; A + 10. \tag{81}$$

Set

$$m_0 \;:=\; \lfloor L^{1/2} \rfloor, \qquad y_j \;:=\; \exp\Big(\tfrac{T_{m_0+j-1}}{B}\Big) \quad (j \ge 1), \tag{82}$$

and

$$R \;:=\; L - m_0 - 4. \tag{83}$$

Note that $R = L - O(L^{1/2}) = (1 - o(1))L$.

Lemma 7.2 (Scale property for H)**.** *For all sufficiently large L and all $1 \le j \le R$,*

$$\exp(y_j^A) \;\le\; y_{j+1}. \tag{84}$$

Proof. Set $m := m_0 + j - 1$. Then

$$y_j = \exp(T_m/B), \qquad y_{j+1} = \exp(T_{m+1}/B).$$

The desired inequality is equivalent to

$$y_j^A \;\le\; \log y_{j+1} = \frac{T_{m+1}}{B},$$

or, since $T_{m+1} = \exp(T_m)$,

$$\exp\Big(\frac{AT_m}{B}\Big) \;\le\; \frac{\exp(T_m)}{B}.$$

Taking logarithms, this is

$$\frac{AT_m}{B} \;\le\; T_m - \log B,$$

equivalently

$$T_m\Big(1 - \frac{A}{B}\Big) \;\ge\; \log B.$$

Because $B = A + 10$, we have $1 - A/B = 10/B > 0$. Also $m \ge m_0$ for every $1 \le j \le R$, and $m_0 = \lfloor L^{1/2} \rfloor \to \infty$. Hence $T_m \to \infty$ uniformly in $1 \le j \le R$, so the displayed inequality holds for all sufficiently large L. □ □

7.2. **The greedy chain construction in the product model.**

Lemma 7.3. *Work in the independent prime model on primes r up to $\exp(y_R^{A-1})$. Then with probability $1 - o(1)$, there is a chain of integers*

$$1 = d_1 < d_2 < \cdots < d_{R+1}, \qquad d_{j+1} \equiv 1 \pmod{d_j}, \qquad d_{j+1} \le y_{j+1}, \tag{85}$$

such that each d_j ($j \ge 2$) is a squarefree product of selected primes lying in the disjoint windows $\big(\exp y_{j-1}, \exp(y_{j-1}^{A-1})\big]$.

Proof. We construct the chain recursively and make deterministic choices. Set $d_1 := 1$. Suppose that d_j has already been constructed, $d_j \le y_j$, and d_j is determined by the selections in the earlier windows $W_1, \dots, W_{j-1}$.

At stage j, use only primes in the fresh window

$$W_j \;:=\; \big(\exp y_j,\, \exp(y_j^{A-1})\big]. \tag{86}$$

Call a squarefree product e *admissible at stage j* if all its prime factors are selected primes from W_j and

$$e \equiv 1 \pmod{d_j}, \qquad e > d_j, \qquad e \le \exp(y_j^A).$$

If at least one admissible e exists, define d_{j+1} to be the least such admissible integer; otherwise the construction fails at stage j.

At this point d_j is measurable with respect to the selections in the earlier windows $W_1, \dots, W_{j-1}$. The fresh window W_j is disjoint from all earlier windows, so conditional on the past and on the realized value of d_j, the selection variables for primes in W_j still have the independent Bernoulli law required by Lemma 6.5. Since $d_j \le y_j$ and Lemma 6.5 is uniform for all $1 \le d \le y_j$, it gives

$$\mathbb{P}(\text{stage } j \text{ fails} \mid \text{past}) \ll y_j^{-c}.$$

The deterministic choice of the least admissible e ensures that d_{j+1} is measurable with respect to the selections in $W_1, \dots, W_j$.

On success, Lemma 6.5 guarantees

$$d_{j+1} \equiv 1 \pmod{d_j}, \qquad d_{j+1} > d_j, \qquad d_{j+1} \le \exp(y_j^A).$$

By Lemma 7.2, valid for every $1 \le j \le R$, $\exp(y_j^A) \le y_{j+1}$, so $d_{j+1} \le y_{j+1}$.

The windows $W_1, W_2, \dots, W_R$ are pairwise disjoint. Indeed, Lemma 7.2 gives $y_{j+1} \ge \exp(y_j^A) > y_j^{A-1}$, hence $\exp(y_j^{A-1}) < \exp(y_{j+1})$, so the upper endpoint of W_j lies below the lower endpoint of W_{j+1}. Thus the primes used at stage j are independent of all previous choices.

Let $\mathcal{F}_j$ be the event that stage j fails, and let $\mathcal{S}_{j-1}$ be the event that all previous stages have succeeded. The estimate above gives, uniformly in the realized past on $\mathcal{S}_{j-1}$,

$$\mathbb{P}(\mathcal{F}_j \mid \text{past}) \ll y_j^{-c}.$$

Therefore

$$\mathbb{P}(\mathcal{F}_j \cap \mathcal{S}_{j-1}) = \mathbb{E}\big[\mathbf{1}_{\mathcal{S}_{j-1}} \mathbb{P}(\mathcal{F}_j \mid \text{past})\big] \ll y_j^{-c}.$$

Thus summing over j is legitimate even though the modulus d_j is random and depends on the earlier stages.

Finally, by the union bound,

$$\mathbb{P}(\text{some stage fails}) \le \sum_{j=1}^{R} O(y_j^{-c}). \tag{87}$$

For all sufficiently large L, the sequence y_j grows at least quadratically: $y_{j+1} \ge y_j^2$. Hence

$$\sum_{j=1}^{R} y_j^{-c} \le \sum_{j \ge 1} y_1^{-c\,2^{j-1}} \le 2\, y_1^{-c} = o(1),$$

because $y_1 = \exp(T_{m_0}/B) \to \infty$. Therefore the construction succeeds with probability $1 - o(1)$. □ □

7.3. **Proof of Theorem 7.1.** The construction in Lemma 7.3 only uses primes up to

$$P := \exp\big(y_R^{A-1}\big) = \exp\Big(\exp\Big(\tfrac{(A-1)T_{m_0+R-1}}{B}\Big)\Big). \tag{88}$$

Now $m_0 + R - 1 = L - 5$, so

$$\log\log P = \frac{(A-1)T_{L-5}}{B}.$$

Since $B = A + 10$, we have $(A-1)/B < 1$. Hence, for all sufficiently large L,

$$\log\log P = \frac{(A-1)T_{L-5}}{B} < T_{L-5}. \tag{89}$$

Exponentiating twice gives

$$\log P < T_{L-4}, \qquad P < T_{L-3}.$$

By (61), $\log x > T_{L-2} \gg T_{L-3}$, so

$$P \;\le\; T_{L-3} \;=\; o(\log x). \tag{90}$$

Set $M := \prod_{p\le P} p$. By Chebyshev (Lemma 2.6), $\log M \le C_\theta P = o(\log x)$, hence $M = x^{o(1)} = o(x)$.

We also record explicitly that the constructed divisors lie below x. For $1 \le j \le R$,

$$d_{j+1} \le y_{j+1} \le y_{R+1} = \exp\Big(\frac{T_{m_0+R}}{B}\Big) = \exp\Big(\frac{T_{L-4}}{B}\Big) < T_{L-3} < T_{L-2} < \log x < x$$

for all sufficiently large L, using (61). Combined with Remark 2.8, this shows that after CRT transfer each constructed d_j is a genuine divisor of the realized integer $n \le x$, not merely a formal product in the independent model.

The event "the greedy chain $\mathcal{C}_H$ of Lemma 7.3 exists" depends only on which primes $p \le P$ divide n. More explicitly, every window W_j used in the construction lies below P, every admissible divisor d_{j+1} is a squarefree product of selected primes from such a window, and the rule choosing the least admissible product is deterministic once the indicators $\big(\mathbf{1}_{p|n}\big)_{p\le P}$ are known. No information about primes above P, and no information about higher powers of primes, is used.

Although the event is phrased in terms of the existence of admissible squarefree products, it is a finite event once the cutoff P is fixed: each window W_j contains only primes $\le P$, and the admissible products are subsets of this finite set of primes. The deterministic rule choosing the least admissible product therefore makes the entire construction a well-defined function of the finite vector $\big(\mathbf{1}_{p|n}\big)_{p\le P}$.

Hence by Lemma 2.7 its density among $n \le x$ equals $\mathbb{P}_{\mathrm{prod}}(\mathcal{C}_H) + O(M/x) = 1 - o(1)$. Whenever $\mathcal{C}_H$ holds, the constructed divisors

$$1 = d_1 < d_2 < \cdots < d_{R+1}$$

all divide n and satisfy $d_{j+1} \equiv 1 \pmod{d_j}$. Thus $H(n) \ge R+1$, and in particular

$$H(n) \;\ge\; R \;=\; (1-o(1))L.$$

This proves (80). □

8. Proof of Theorem 1.1

Combine Theorems 3.1, 4.1, 5.1, and 7.1. Thus, with $L := \log_* x$, for all but $o(x)$ integers $n \le x$,

$$\big(\tfrac{1}{2} - o(1)\big)L \;\le\; h(n) \;\le\; \big(\tfrac{1}{2} + o(1)\big)L, \tag{91}$$

$$\big(1 - o(1)\big)L \;\le\; H(n) \;\le\; \big(1 + o(1)\big)L. \tag{92}$$

Here and below we have intersected the four density-one sets supplied by those theorems; the union of their exceptional sets still has size $o(x)$.

It remains to replace $L = \log_* x$ by $\log_* n$. We first record a uniform comparison on the interval $[x^{1/2}, x]$. Since $\log_* x = L$, we have

$$T_{L-1} < x \le T_L$$

for $L \ge 1$. Hence, for $n \in [x^{1/2}, x]$,

$$n \;\le\; x \;\le\; T_L,$$

so $\log_* n \le L$. Also

$$n \;\ge\; x^{1/2} \;>\; T_{L-1}^{1/2} \;=\; \exp(T_{L-2}/2).$$

For all sufficiently large L, $\exp(T_{L-2}/2) > T_{L-2}$. Therefore $n > T_{L-2}$, and so $\log_* n \ge L-1$. Consequently,

$$\log_* n = L + O(1) \tag{93}$$

uniformly for $n \in [x^{1/2}, x]$.

This comparison is uniform on $[x^{1/2}, x]$. Hence any estimate of the form $A(n) = cL + o(L)$ holding outside an exceptional set of size $o(x)$ may be rewritten on the same density-one set, after discarding $n \le x^{1/2}$, as

$$A(n) = \big(c + o(1)\big) \log_* n,$$

with the same constant c.

The integers $n \le x^{1/2}$ form an exceptional set of size $O(x^{1/2}) = o(x)$. Discarding them, (93) gives

$$L = (1 + o(1)) \log_* n$$

uniformly on the remaining density-one set. Substituting into (91) and (92) yields

$$h(n) = \Big(\tfrac{1}{2} + o(1)\Big) \log_* n, \qquad H(n) = \big(1 + o(1)\big) \log_* n$$

for all but $o(x)$ integers $n \le x$. Finally,

$$\frac{H(n)}{h(n)} \;=\; \frac{\big(1 + o(1)\big) \log_* n}{\big(\frac{1}{2} + o(1)\big) \log_* n} \;=\; 2 + o(1),$$

because the lower bound for $h(n)$ ensures $h(n) \to \infty$ on the same density-one set. This proves (3)–(5). □

9. Concluding remarks

9.1. **What is and is not proven.** We have proven the *normal orders* of $h(n)$ and $H(n)$, i.e. asymptotic statements about a density-one set of n. Theorem 1.1 does *not* assert that $h(n) \le (\frac{1}{2} + o(1)) \log_* n$ pointwise, nor that $H(n) \ge (1 - o(1)) \log_* n$ pointwise; both pointwise statements are easily seen to fail (e.g. for n a product of a long prime chain $p_1 \equiv 1 \pmod{p_0}, p_2 \equiv 1 \pmod{p_1}, \dots$). Erdős' original belief that $h(n)$ has a normal order $\log_* n$ is *at least partially* correct: we get $\Theta(\log_* n)$ with the precise constant $1/2$.

9.2. **Sharpness of the second-order term.** Our bounds carry an additive slack of order $L^{1/2}$, where $L = \log_* x$, coming from the small-prime cutoff $Y = \lfloor L^{1/2} \rfloor$. After replacing L by $\log_* n$ on the final density-one set, this is an additive $O((\log_* n)^{1/2})$ slack. The truth is presumably $h(n) = \frac{1}{2} \log_* n + O(\log \log_* n)$ a.a., but proving this would require a finer treatment of bad pairs at moderate scales. We have not attempted it here.

9.3. **Convergence in probability.** The methods of Sections 5 and 7, together with the upper bounds, show convergence in probability:

$$\frac{h(n)}{\log_* n} \to \frac{1}{2}, \qquad \frac{H(n)}{\log_* n} \to 1$$

with respect to natural density. We do not claim L^p-convergence here; that would require additional uniform-integrability estimates beyond the arguments of this paper.

9.4. **Why the constant is $1/2$ and not, say, 1.** The factor of 2 between the prime-chain and divisor-chain rates comes from the fact that two prime steps are needed to pass through one full "U-tower" level, while one divisor step suffices. The bad-pair analysis is perfectly parallel for h and H, but the cost-per-step asymmetry — which can be traced to the Brun–Titchmarsh-controlled scale $Q(p) = \exp\big(\exp\big(p/(\log p)^2\big)\big)$ versus the pure exponential scale $\exp \sqrt{d}$ — forces the factor of 2.

9.5. **Possible refinements.**

(a) *Sharper constants in second order.* One can probably push the additive $O(L^{1/2})$ error in (51), equivalently $O((\log_* n)^{1/2})$ on the final density-one set, down to $O(\log\log_* n)$ by being more careful with the bad-pair tail. We have not explored this.

(b) *Distributional questions.* It is natural to ask whether, after centering and normalizing, $\big(h(n) - \frac{1}{2}\log_* n\big)$ has a limiting distribution. This appears to be a substantially harder question: the prime-chain construction in Section 5 is essentially a renewal process, but the Siegel–Walfisz error term is too crude to give a CLT directly.

(c) *Worst-case bounds.* The pointwise extremal problem is quite different from the normal-order problem considered here. For instance, if $n = \operatorname{lcm}(1, 2, \ldots, k)$, then

$$1 < 2 < 3 < \cdots < k$$

is a valid divisor chain, since $j + 1 \equiv 1 \pmod{j}$, and therefore $H(n) \geq k$ while $\log n \sim k$ by the prime number theorem in the form $\log \operatorname{lcm}(1, \ldots, k) \sim k$. Similarly, special integers built from long prime chains can make $h(n)$ much larger than its normal order. These pointwise phenomena are far from the density-one behavior proved here.

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